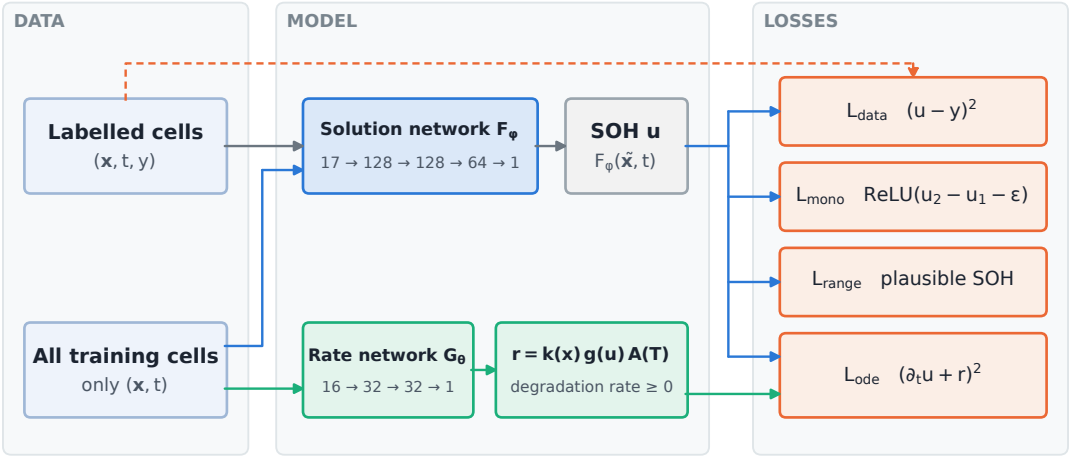

$$L = L_{data} + w(s)\,(\alpha L_{ode} + \beta L_{mono} + \gamma L_{range}) \qquad \text{joint update of } \varphi, \theta, \lambda, E_a$$